

Deliverable D15 · Work Package WP4

Written Summary of the Research – Report - Seminar

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Principal investigator: Ing. Roman Budjač, PhD.

Written Summary of the Research This document is the D15 deliverable of Work Package WP4 of the SAMQ-NN project and presents a coherent written summary of the finalized method and research results. Unlike Deliverable D13, which provides a technical proof of concept in the form of code and experimental graphs, this document presents the method and its validation in readable prose that is understandable even without direct access to the implementation. The goal is to summarize what was proposed, how it was validated, and what the benefits and limitations of the resulting solution are.

1. Introduction

Quantization of neural networks reduces the model's memory and computational requirements by representing weights with lower bit precision than the standard 32-bit floating-point format. However, this savings may come at the cost of reduced accuracy, and it is therefore important to be able to monitor the quality of quantization and its impact on the model. The SAMQ-NN project proposes a statistical approach for this purpose: monitoring the extent to which the distribution of weights in the quantized model deviates from that of the original, full model. Work Package WP3 implemented this approach as a new training strategy, and Work Package WP4 is experimentally validating it. This summary builds on the aforementioned work and synthesizes it into a comprehensive overview.

2. The Proposed Method

At the core of the method is a similarity index between the reference weight distribution of the full-precision (FP32) model and the current weight distribution of the quantized model. Both distributions are represented by histograms based on the same, predefined class boundaries, and are compared using statistical measures. The Jensen–Shannon divergence is used as the primary measure; it is symmetric and bounded on the interval from zero to one, making it suitable for thresholding and automated decision-making. It is supplemented by the Kullback–Leibler divergence, which is sensitive to information loss, and cosine similarity, which captures the orientation of weight vectors; Wasserstein distance is also available as an extension.

The index is not calculated only after quantization is complete, but continuously during training, via a custom callback class in the Keras framework that runs at the end of each epoch. Both

the reference distribution and the class boundaries remain unchanged throughout training, ensuring that every comparison is consistent. Quantization is performed on a per-channel basis, which has proven essential for the robustness of deep networks. The index can be evaluated globally across the weights of the entire model or layer by layer, thereby identifying the most sensitive layer.

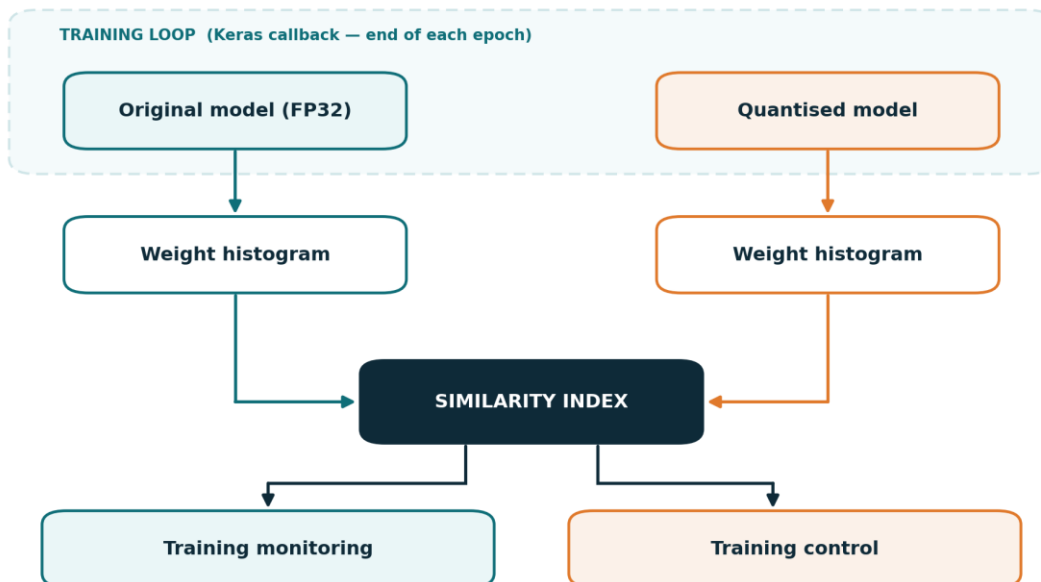


Fig. 1 Method overview: a similarity index between the FP32 weight distribution and the quantized model, calculated in real time during training via a Keras callback.

3. Verification Method

The method was validated on standard benchmark datasets—Digits, Fashion-MNIST, MNIST, CIFAR-10, and the ImageNet-V2 validation subset—and on architectures ranging from a small convolutional network to pre-trained MobileNetV2 and ResNet50 models. The evaluation tracked accuracy (top-1, and top-5 for ImageNet-V2) and the loss function value compared to a fully accurate reference, as well as the performance of the index against the four validation criteria specified in the D10 output: determination of the index threshold, correlation of the index with accuracy, stability of the index across epochs, and reproducibility of behavior across datasets and architectures.

4. Key Findings

Experimenty potvrdili navrhnutý koncept. Index monotónne sleduje tvrdosť kvantizácie, Spearmanova poradová korelácia medzi bitovou hĺbkou a Jensenovou–Shannonovou divergenciou dosahuje naprieč datasetmi približne mínus 0,99. Pri osembitovej kvantizácii ostáva presnosť prakticky bezstratová, a to vrátane modelu ResNet50 na datasete ImageNet, zatiaľ čo agresívna dvojbitová kvantizácia presnosť výrazne zhoršuje. Počas kvantizačne uvedomelého tréningu je index stabilný, s malou smerodajnou odchýlkou naprieč epochami, čo potvrdzuje, že váhy ostávajú reálne kvantizované a tréning ich nevracia do plnej presnosti. Analýza po vrstvách identifikovala prvú konvolučnú a výstupnú vrstvu ako najcitlivejšie na kvantizáciu.

Quantization itself yields real cost savings: the theoretical compression ratio is four times higher with eight bits and eight times higher with four bits compared to the full-precision model,

a finding confirmed by the actual export of the model to integer (int8) format. An important finding is that the relationship between the index and accuracy is nonlinear: the index reacts to a change in the weight distribution before a decline in accuracy becomes apparent, making it an early warning signal rather than merely an additional indicator.

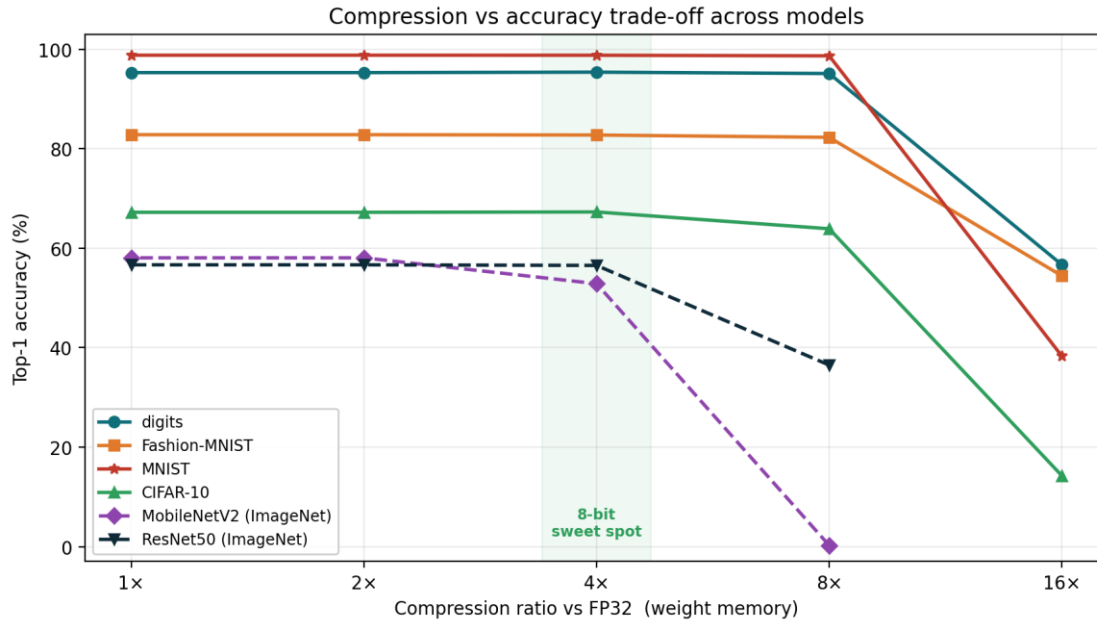


Fig. 2 A trade-off between compression and accuracy across models (digits, Fashion-MNIST, MNIST, CIFAR-10, MobileNetV2, ResNet50). The x-axis represents the compression ratio of the weight memory (32/bits, log), and the y-axis represents absolute top-1 accuracy. The range around fourfold compression (8 bits) is the “sweet spot.”

Index validation

The optimal bit depth is not determined by the index; it results from a direct measurement of precision and is consistent with established practice: 8-bit quantization is nearly lossless at 4x compression. It is precisely this independently established fact that serves as the criterion for verifying the index. The index remains below the threshold exactly where accuracy is preserved (≥ 8 bits) and increases where accuracy decreases (≤ 4 bits), with a Spearman correlation with bit depth of ≈ -0.99 . The index’s alignment with this reference behavior confirms that it is a useful proxy for quantization quality; it is measured from the distribution of weights but reflects what works in practice. However, the agreement is not unconditional: for the MobileNetV2 model, the weight index fails to capture the accuracy collapse at four bits, which limits the validity of the method. The index is a reliable indicator for the tested family of models, but both the threshold and the predictive power itself (weights versus activations) must be calibrated for a given architecture.

Selecting the bit depth (6 vs. 8 bits). When an index-driven strategy minimizes the bit depth while keeping the index below the threshold, it naturally chooses approximately 6 bits; at this width, the index is well below the threshold ($JS \approx 0.001$) and accuracy is preserved, making 6 bits the analytically most efficient and, across the tested datasets, a nearly lossless choice. However, the recommended target for deployment remains 8 bits (INT8): integer hardware and runtime environments, mobile and edge NPUs, INT8 accelerators, TFLite/ONNX integer cores natively support 8 bits, whereas 6 bits is rarely supported natively by hardware and, in practice, would be stored as 8 bits anyway, thereby negating the memory advantage during

runtime. The method therefore offers both: the analytical optimum (≈ 6 bits) and a practically deployable standard (8 bits), which is key for deployment.

Table 1 summarizes the main numerical results for individual datasets and architectures.

Dataset / model	FP32 top-1	8-bit top-1	4-bit top-1	$\rho(\text{bity, JS})$
digits	0,967	0,966	0,964	-0,984
Fashion-MNIST	0,907	0,907	0,900	-0,984
MNIST	0,987	0,987	0,987	-0,984
CIFAR-10	0,638	0,637	0,601	-0,984
MobileNetV2 (ImageNet)	0,573	0,519	0,002	-1,000
ResNet50 (ImageNet)	0,563	0,562	0,365	-1,000

Tab. 1 Accuracy (top-1) at selected bit depths and Spearman's correlation between the index and bit depth. Eight-bit quantization is practically lossless; at four bits, the decline depends on the architecture (MobileNetV2 collapses, while ResNet50 degrades more moderately).

The table shows that 8-bit quantization is virtually lossless across all datasets—the drop in top-1 accuracy is within the margin of error between runs, including for the ResNet50 model on ImageNet. At four bits, performance begins to depend on the architecture: small convolutional networks and ResNet50 generally maintain their accuracy, but MobileNetV2 drops to nearly zero, even though its weight index remains negligible—that is, the degradation originates outside the weight distribution and is not captured by the global weight index. The $\rho(\text{bits, JS})$ column confirms that the index monotonically tracks the severity of quantization (≈ -0.98 for small models, -1.00 on ImageNet).

Table 2 summarizes the trend across all tested bit depths (average of small datasets).

Bit width	Average decline top-1	Average increase in losses	Average Index (JS)	Compression ratio
32	0,0 %	1,00×	0,0000	1×
16	0,0 %	1,00×	0,0000	2×
8	0,0 %	1,00×	0,0001	4×
4	-1,7 %	1,11×	0,0819	8×
2	-63,5 %	20,33×	0,6322	16×

Tab. 2 Average decrease in accuracy (top-1), average increase in the loss function (as a multiple of the FP32 value), and average index (JS) by bit depth across small datasets (digits, Fashion-MNIST, MNIST, CIFAR-10). The accuracy, loss, and index values are measured from training; the compression ratio (32/bits) is theoretical. The large increase in loss at two bits is partly due to datasets with very low FP32 error (e.g., MNIST), where even a small absolute change results in a large multiple.

The table shows a common trend across the four small datasets: up to eight bits, both the index and accuracy remain unchanged (fourfold compression at virtually no cost). At four bits, the index increases noticeably, while accuracy drops only slightly ($\approx 1.7\%$) and loss increases by about 11%; thus, the index rises before the degradation fully manifests itself in performance. With two bits, a collapse occurs: the index approaches its upper limit, accuracy drops by more than half, and the error function increases by an order of magnitude. It is precisely this nonlinear lead time that makes the index an early warning signal, not merely an additional indicator.

Cross-dataset strategy (SMT vs. baseline)

The proposed strategy (SMT) automatically selects the bit depth for each dataset based on the index threshold and monitors the per-layer index during quantization-aware training, with

the option for early termination. To demonstrate its true benefit, we compare it with a naive baseline, fixed 8-bit quantization for the same number of epochs, without monitoring and without automatic selection. The results are summarized in Table 3.

Dataset	SMT bity	SMT epochy	SMT top-1	Baseline (8-bit) top-1	Kompresia (SMT / baseline)
digits	6	4	0,972	0,978	5,33× / 4×
Fashion-MNIST	6	4	0,911	0,902	5,33× / 4×
MNIST	6	30	0,991	0,992	5,33× / 4×
CIFAR-10	6	5	0,649	0,629	5,33× / 4×

Tab. 3 Strategy (SMT) versus a fixed 8-bit baseline with the same epoch limit. SMT selects the bit depth automatically and uses index-controlled early termination; the selected depth, the number of epochs (30 = early termination did not trigger within the budget), and the achieved accuracy are reported. The compression ratio is theoretical (32/bits).

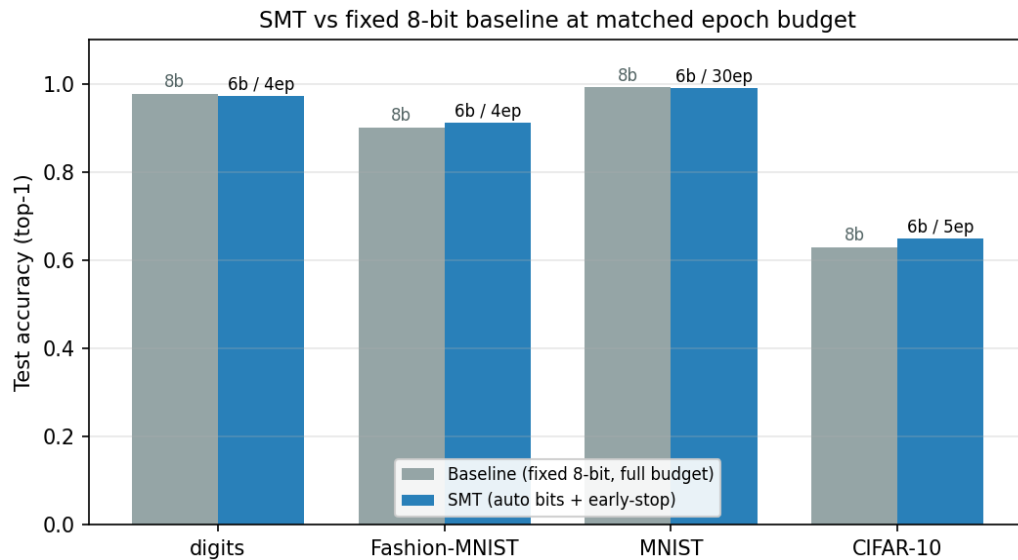


Fig. 3 Accuracy (top-1) of the SMT strategy (automatically selected ≈ 6 bits + early termination) compared to a fixed 8-bit baseline with the same number of epochs. The labels indicate the selected bit depth and the number of epochs.

Three findings emerge from the comparison. First, across all four datasets, the strategy consistently selects approximately six bits (5.33× compression), thereby outperforming the 8-bit baseline (4×) in terms of memory savings. Second, this savings does not come at the expense of accuracy: on all four datasets, SMT’s accuracy is comparable to that of the 8-bit baseline, and the differences are small (ranging from -0.6 to $+2.0$ percentage points) and within the variability between runs, so the 6-bit approach is virtually lossless across the tested datasets. Third, index-driven early stopping was triggered during a full run on three of the four datasets (digits, Fashion-MNIST, and CIFAR-10 as early as the 4th–5th epoch; MNIST reached the 30-epoch limit), thereby significantly reducing the number of training epochs without any loss of accuracy.

Early termination as an index stabilization detector

The mechanism stops training when the per-layer index stabilizes—that is, when the weight distribution relevant for quantization stops changing. It has an important safety feature: it is

triggered only upon stabilization, so it never stops prematurely; if it is not triggered, training simply runs until the specified limit is reached, and nothing is lost. At full speed, the index stabilized quickly on most datasets, and the mechanism thus saved a significant portion of the training time without any loss of accuracy. For digits, Fashion-MNIST, and CIFAR-10, training ended as early as the 4th–5th epoch instead of 30 (e.g., digits 0.972 vs. 0.978 for the baseline), while on MNIST, the index did not stabilize within the budget and training ran to the ceiling. We therefore report the stabilization epoch as a measure of the speed at which quantization stability is achieved, not as a failure; the mechanism is recommended for use in conjunction with an epoch cap and accuracy monitoring, not on its own.

5. Conclusion, Benefits, and Limitations

Based on the validation results, the method was revised and refined into its current form. Quantization was standardized to a channel-by-channel scheme (rather than tensor-by-tensor), as it more faithfully preserves the weight distribution and yields a more stable index; the threshold value ($\tau = 0.01$) was calibrated against independently measured losslessness and is understood as an architecture-dependent parameter, not a universal constant. The metric itself was reevaluated and reframed from a “detector” to a proxy validated against established practice, thereby explicitly defining its scope of validity. The main impetus for further refinement is the observed limitation of the weight index when degradation occurs outside the weight distribution (MobileNetV2), which leads to the extension of monitoring to include activations.

The main advantage of this solution is that the index is computationally inexpensive, and its overhead is negligible compared to a single forward and backward pass through the network. It can be computed on the fly during training; in fact, multiple indices can be computed simultaneously. The index is dataset-agnostic and independent of the choice of a specific metric, since the individual measures are strongly correlated with one another. It can be used in two modes: as a predictor to select an appropriate bit depth before training, and as a monitor to track the model’s status during training, including early termination. Index-guided early termination works and has been implemented in full on three out of four datasets (digits, Fashion-MNIST, and CIFAR-10), but it may not work everywhere; for example, on MNIST, the index did not stabilize within the epoch budget and training continued to completion. It is important to use it in conjunction with an epoch cap and accuracy checks, not on its own.

The validation also included defining the method’s scope of validity. The index specifically tracks the distribution of weights and therefore most reliably captures degradation that manifests in the weights. At the same time, the experiments made it possible to identify cases where the decline in accuracy stems from factors outside the weight distribution—for example, in the MobileNetV2 model with 4-bit quantization, where the degradation occurs primarily at the activation level. This finding precisely delineates the scope for which the weight index is sufficient and naturally defines its extension to include activation monitoring. For robust performance in deep networks, channel-wise quantization is used, and the index threshold is calibrated for a given architecture; the index is therefore proposed to be monitored alongside accuracy, thereby providing a complete picture of the relationship between changes in the weight distribution and model performance.

6. Next Steps

The research has demonstrated that the proposed statistical similarity index is a practical and cost-effective tool for monitoring quantization directly during training, which meets the

validation criteria set forth in Output D10 and fulfills the project's objective: to validate the concept on standard datasets and evaluate its benefits and limitations. The future direction of the research stems directly from the findings and the defined scope of the method: extending monitoring from weights to activations, proposing a per-layer index-guided mixed accuracy strategy, and performing a more thorough calibration of threshold values. The results summarized in this document form the basis for a forthcoming scientific publication (D16) and for the project's final report (D17).

Ing. Roman Budjač, PhD.